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2.14 Active Transport

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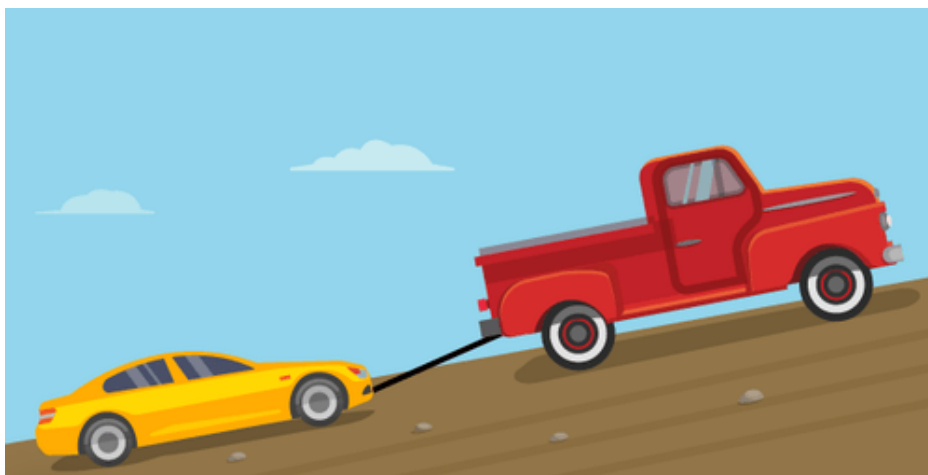
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Assign to class

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Difference between active and passive transport



[Figure 1]

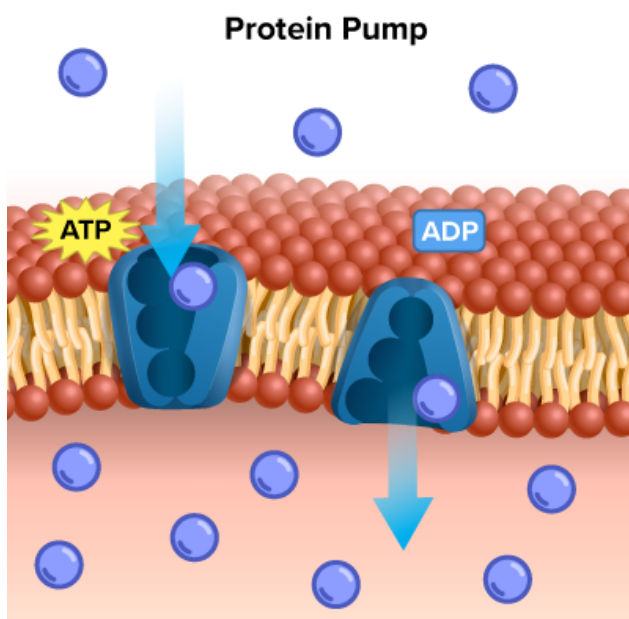
need to move something really heavy?

by using energy.

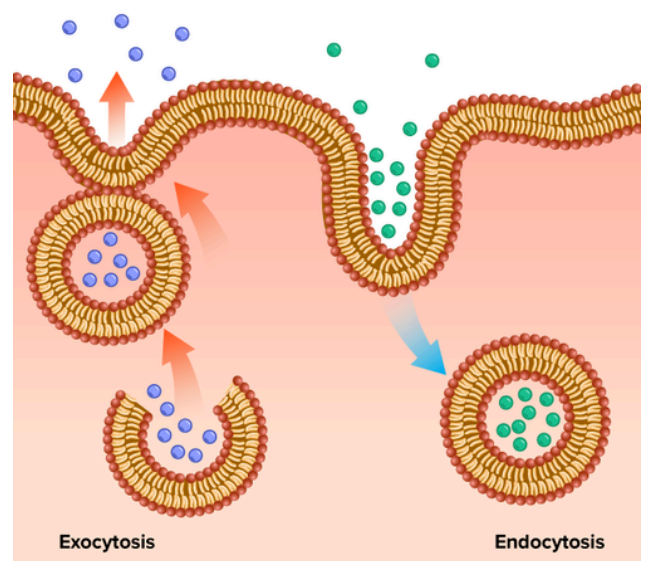
Active Transport

In contrast to passive transport, which does not require energy and carries molecules or **ions** down a concentration gradient, active transport pumps molecules and ions against a concentration gradient. Sometimes an **organism** needs to transport something against a concentration gradient. The only way this can be done is through active transport, which uses energy in the form of ATP (adenosine triphosphate) that is produced by **cellular respiration**. In active transport, the particles move across a **cell membrane** from a lower concentration to a higher concentration (see the **Figure below**). **Active transport** is the energy-requiring process of pumping molecules and ions across membranes "uphill" - against a concentration gradient.

- The active transport of small molecules or ions across a cell membrane is generally carried out by **transport proteins** that are found in the membrane, also known as a **protein pump**.
- Larger molecules such as **starch** can also be actively transported across the cell membrane by **vesicle transport** processes called **endocytosis** (substances enter the cell) and **exocytosis** (substances exit the cell).



[Figure 2]



[Figure 3]

Active Transport

Teacher's Pet



Watch on

Homeostasis and Cell Function

Homeostasis refers to the **balance**, or equilibrium, within the cell or a body. It is an organism's ability to keep a constant internal environment. Keeping a stable internal environment requires constant adjustments as conditions change inside and outside the cell. The adjusting of systems within a cell is called homeostatic regulation. Because the internal and external environments of a cell are constantly changing, adjustments must be made continuously to stay at or near the set point (the normal level or range). Homeostasis is a dynamic equilibrium rather than an unchanging state.

CAN YOU MATCH?

Try matching the type of active transport with their functions?





Plant cells use active transport to transport materials throughout the plant. Transport of ions and **minerals** from the **soil** to **roots**, transport of sugars from leaves to **fruit**, transport of nitrate and chloride from the **cytosol** to **vacuole** are some examples of active transport in plants.

Summary

- Active transport is the energy-requiring process of pumping molecules and ions across membranes against a concentration gradient.
- Active transport processes help maintain homeostasis.

Review

1. What is active transport?
2. Explain how cell transport helps an organism maintain homeostasis.



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